Twelve-variable Diophantine forms of ZFC consistency

Monic elimination, signed domains, and smaller fixed coefficients

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Abstract

We combine the existing blank-input K215 compiler, weighted square-root elimination, and signed-domain arguments with a new elimination based on algebraic integers. The resulting explicit polynomial has twelve positive integer unknowns, degree 1529, and a division-free evaluator with 117 binary arithmetic operations. Its three-equation presentation has maximum degree 1520 and costs 112 operations. An intermediate thirteen-variable polynomial has degree 319, improving the reviewed thirteen-variable degree 515. A signed version uses 26 unrestricted integer unknowns, improving 29. A separate combination gives a 59-operation, seven-equation system in sixteen positive unknowns. Finally, a strict four-cell compiler uses 5,252 tiles and base $2^{5^{16}}$, reducing the binary length of the base-minus-one coefficient by a factor of 25 from the reviewed T4 recipe. All constructions have solutions exactly when K215 halts on its blank tape, and hence their unsolvability is equivalent in ZFC to syntactic Con(ZFC). The new mathematics is proved here and accompanied by finite, reproducible checks. It is not a new Lean or Rocq formalization and makes no global minimality claim.

1 Results, comparison, and the precise claim

1.1 The outcome

The main new equation is

$$\boxed{\mathcal{P}_{12} = N_k N_t(1 + \mathcal{R}_{12}) - 1 = 0,} \tag{1}$$

in the twelve positive integer unknowns

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, u. \tag{2}$$

Every symbol on the right of (1) is an explicitly defined polynomial abbreviation. Section 3 defines them, including a division-free formula for $\mathcal{R}_{12}$. Four fixed integer coefficients are specified by a terminating recipe in Section 8. There is no unspecified machine index, no unknown coefficient, and no variable exponent in the equation.

Theorem 1.1. *For the fixed coefficient tuple of Section 8,*

$$\mathrm{ZFC} \ \vdash \ [\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{12}\, \mathcal{P}_{12}(\boldsymbol{x}) = 0] \ \longleftrightarrow \ \mathrm{Con(ZFC)}.$$

The corresponding three-equation system $N_k = N_t = 1$, $\mathcal{R}_{12} = 0$ has the same existence condition. The polynomial in 26 unrestricted integers defined in Section 6 has the analogous equivalence over $\mathbb{Z}$.

The proof is a composition of ordinary mathematical arguments formalizable in ZFC. In particular, it does not infer an internal ZFC theorem merely from the soundness of a proof assistant. The inherited K215 calibration, arithmetic compiler, and new algebraic reductions all admit the usual finite coding within set theory. The machine endpoint has existing Lean and Rocq evidence; the new Diophantine bridge does not acquire that formal status by composition.

1.2 A comparison with the reviewed snapshot

The review is pinned to `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`. This snapshot contains the seven wave-2 packages numbered 09–15. Their reconciliation note is included as `reference/comparison_at_pin.md`; its statistics are the packages' stated figures, not new independent rebuilds of all seven packages. The table compares matching objectives rather than declaring one universal notion of simplicity.

Construction	Unknowns	Eqs.	Degree	Ops.	Domain
Reviewed 09 low weight	13	1	515	85	positive
New main	**12**	1	1529	117	positive
New main system	12	3	1520	112	positive
New intermediate	13	1	**319**	91	positive
New intermediate system	13	4	288	83	positive
Reviewed integer best (14)	29	1	188	121	unrestricted $\mathbb{Z}$
New integer form	**26**	1	2993	169	unrestricted $\mathbb{Z}$
Reviewed retained system (09)	16	7	24	60	positive
New retained system	16	7	24	**59**	positive
Reviewed minimum-ops polynomial	15	1	155	67	positive

Here "degree" for a system means the maximum total degree of its residual polynomials. An operation is a binary $+$, $-$, or multiplication, including multiplication by a fixed numeral. Reusing an already computed register is free; supplying a fixed coefficient is free; equality tests are not counted as arithmetic gates. Polynomial abbreviations do not introduce unknowns. All new circuit counts have been checked on the emitted live gate arrays.

The twelve-variable result improves the reviewed explicit variable count, at the cost of higher degree and more arithmetic. The thirteen-variable result improves degree at the same variable count. The 59-operation result improves the system count. We do not find a polynomial below 67 operations. The existing 32-variable quadratic lift, degree-22 fifteen-variable systems, and other low-degree alternatives remain useful for their respective objectives.

There is also a separate coefficient improvement:

Recipe	Tiles	Base B_{base}	Bits in $B_{\text{base}} - 1$
Reviewed T4 (11)	37,075	$2^{5^{18}}$	$3,814,697,265,625$
New strict four-cell	5,252	$2^{5^{16}}$	$152,587,890,625$

Even the new coefficients are enormous. They are specified exactly, not expanded. The bit count for $B_{\text{base}} - 1$ is about 19.1 decimal GB; the coefficient recipe is the practical representation.

1.3 What was combined

Package 09 contributes weighted elimination of the strong, first, and auxiliary Pell roots. Package 12 supplies a signed main-root and auxiliary-quotient interface, together with the transport-coordinate shift. Package 14 shows how to retain only seven positive coordinates by choosing an odd multiplier and a square power scale. Package 11 supplies the four-cell local language, native compiler layout, and a much smaller coefficient tuple.

The new step removes the main congruence quotient together with the old combined-root coordinate. A monic polynomial makes the replacement root an algebraic integer. A short norm argument forces it to be rational and hence integral. A separate Pell lemma proves that the remaining rational auxiliary ordinate is automatically integral. These two facts are what make the resultant elimination sound; merely taking a resultant would not suffice.

Classical universal constructions can use fewer unknowns: the repository's formal development includes an existence theorem for nine unknowns, while eleven-variable integer representations are discussed in modern work [4, 6]. Our claim is a smaller concrete specialization of the reviewed native-compiler constructions with explicit coefficients, equations, degree, and evaluator. It is not a new universal variable bound.

2 The inherited arithmetic interface

Write the four fixed coefficients as

$$\beta = B_{\mathrm{base}} - 1, \qquad K_0 = D_C + B_{\mathrm{base}} D_R, \qquad M_C, \qquad M_F^* = M_F + B_{\mathrm{base}} - 1.$$

The star is essential: the polynomial uses the shifted mask, not M_F. The compiler establishes

$$0 < M_C, M_F < B_{\mathrm{base}} - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_F \equiv 4 \pmod 8,$$
$$\mathrm{pc}(M_C) + \mathrm{pc}(M_F) = \log_2 B_{\mathrm{base}}.$$

All four ports must be taken from one whole authenticated recipe.

For positive supplied coordinates, define

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & & \\
G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F^*)J, & & \\
L_0 &= E(kY), & A &= 1 + L_0(L_0 + k), & & \\
N_k &= k - hE - R, & N_t &= (K_0 + w)C + (q - F) - z(q-1). & &
\end{aligned} \tag{3, 4}$$

The letter A here denotes the first Pell radicand. The usual Pell sequence parameter is $a + 2$, which we will sometimes call A_0.

The normalized fifteen-coordinate interface restores four further witnesses f, τ, y, γ and has equations

$$\tau^2 = A, \tag{5}$$
$$(X + ac + H\gamma)^2 - \Delta c^2 = 1, \tag{6}$$
$$f^2 - \Delta i^2 c^4 = 1, \tag{7}$$
$$QV_f^2 - (Q-1)y^2 = 1, \tag{8}$$
$$N_k = N_t = 1, \tag{9}$$

where $Q = (\Delta i c^2)^2$ and $V_f = c(Tf - 1) - Rf^2$. There are fifteen supplied coordinates before the eliminations: the twelve in (2), with u removed and f, τ, y, γ inserted.

The inherited compiler proves that this system has a positive solution exactly when blank K215 halts. Its signed extension permits the appropriate restored roots, T, and γ to be signed. One way to apply the ordinary strong-norm version to (7) is to put

$$i_{\text{ordinary}} = \Delta i. \tag{10}$$

Then $(i_{\text{ordinary}} c^2)^2 = \Delta(f^2 - 1)$, which is exactly its strong equation. The conversion is a proof step; it introduces no new supplied variable.

For reference, the main logical steps in the inherited argument are as follows. Weak transport gives $C \geq 0$ even when $N_k = N_t = -1$. The authentic masks yield

$$q \geq 16, \quad q(q^2-1) < R < q^4 - 2, \quad E \geq q^4 > R + 2, \quad Y \geq 4096. \tag{11}$$

The first and main Pell equations classify $k = 2\psi_{2XY^2+1}(n)$ and $c = \psi_{a+2}(p)$. The strong equation and signed auxiliary congruences recover $p = R$. The no-wrap estimate and a Pell ratio exclude the negative outer unit signs. The main-root congruence then gives $X = 2^R$, with the negative main root excluded by the signed argument. Half-binomial extraction restores the population condition and the native masks. Carry-free native convolution recovers a periodic local computation tableau containing Start. A finite strip around that Start gives a genuine blank halting run.

Conversely, any finite halting run admits independent spatial and temporal padding. Free dummy bits tune its actual packed rank, and the positive Pell converse supplies all remaining coordinates. Detailed inherited proofs are included under `reference/`; the new compiler adapter is proved in Section 8.

3 The twelve-variable equation in full

In addition to (3)–(4), put

$$\begin{aligned}
I &= ic^2, & D &= 1 + \Delta I^2, & Q &= \Delta(D-1), & d &= Q - 1,\\
v_0 &= cT, & b_0 &= c + RD, & & & & \\
P &= QDv_0^2, & Z &= Qb_0^2, & e &= P + Z - 1, & a_0 &= PZ,\\
M &= 2(kI^2 + T^2), & N &= dM^2 A. & & & &
\end{aligned} \tag{12}$$

In particular $Q = (\Delta i c^2)^2 > 1$. Define a monic polynomial in a formal indeterminate v:

$$\begin{aligned}
\mathcal{H}_8(v) = {} & \big[v^4 - 2d(N+e)v^2 + d^2((N-e)^2 + 4a_0)\big]^2 \\
& - 16a_0 d^2 \big[v^2 + d(N-e)\big]^2.
\end{aligned} \tag{13}$$

The indeterminate v is eliminated; it is not a thirteenth unknown. Put

$$B_* = 1 + \Delta c^2, \qquad W = X + ac + Hu, \qquad p = Hc, \tag{14}$$

and let

$$\mathcal{R}_{12} = \mathrm{Res}_v(\mathcal{H}_8(v), (W - pv)^2 - B_*). \tag{15}$$

Equation (1), with these definitions, is the promised polynomial.

3.1 A division-free evaluator

The notation "resultant" in (15) is only a concise specification. The following straight-line formulas evaluate exactly the same integer polynomial using only addition, subtraction, and multiplication:

$$\begin{aligned}
r_0 &= W^2 + B_*, & r_1 &= 2W, & t_0 &= dp^2, & t_1 &= t_0^2,\\
S &= N - e, & \ell &= 2t_0(N + e), & v_c &= t_1(S^2 + 4a_0),\\
v_R &= r_0^2 + B_* r_1^2 - \ell r_0 + v_c, & v_I &= r_1(2r_0 - \ell),\\
u_R &= r_0 + t_0 S, & \lambda &= 16 a_0 t_1,\\
h_R &= v_R^2 + B_* v_I^2 - \lambda(u_R^2 + B_* r_1^2),\\
h_I &= 2(v_R v_I - \lambda u_R r_1),\\
\mathcal{R}_{12} &= h_R^2 - B_* h_I^2.
\end{aligned} \tag{16}$$

They express the exact identity

$$h_R + h_I\sqrt{B_*} = p^8 \mathcal{H}_8((W + \sqrt{B_*})/p).$$

Taking its conjugate product gives (15), since the quadratic's leading coefficient is p^2. No division or square root is used in the actual evaluator. The complete 117-gate source is `data/merger12_product.json`; the equations above are its mathematical specification, not a hidden extra-variable system.

4 Why the elimination is sound

4.1 The monic-root merger

Lemma 4.1. *Let $a, c \in \mathbb{Z}_{>0}$, $\Delta = (a+2)^2 - 1$, $H = 4a + 3$, and $B_* = 1 + \Delta c^2$. Suppose a monic polynomial in $\mathbb{Z}[v]$ has a common root with $(W - Hcv)^2 - B_*$, where $W \in \mathbb{Z}$. Then B_* is a square and the common root is an integer.*

Proof. The common root is $(W \pm \sqrt{B_*})/(Hc)$ and is an algebraic integer by monicity. If B_* is nonsquare, the difference of this root and its conjugate is a nonzero algebraic integer in the quadratic field. Its norm is

$$-\frac{4B_*}{H^2c^2}.$$

But

$$H^2 - 4\Delta = 12a^2 + 8a - 3 \geq 17,$$

so $0 < 4B_* < H^2c^2$. The norm lies strictly between -1 and 0, contradicting the integrality of the norm of an algebraic integer. Thus B_* is square. The root is now rational, and every rational algebraic integer is an integer. □

Applied to (13), this lemma gives an integer v. Consequently

$$\gamma = u - cv \in \mathbb{Z}, \qquad (X + ac + H\gamma)^2 - \Delta c^2 = 1. \tag{17}$$

It is essential that γ may initially be negative. The inherited signed-domain theorem handles that sign; we do not assume the positive main root merely because the original compiler used one.

4.2 A rational Pell ordinate saturates

Lemma 4.2. *If $S \geq 2$ and V are integers and $y \in \mathbb{Q}$ satisfies*

$$S^2V^2 - (S^2 - 1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $S^2 - 1 = b^2 D_0$ with $D_0 > 1$ squarefree. Put $x = S|V|$ and $z = b|y|$. Then $x^2 - D_0 z^2 = 1$. In a reduced rational expression for z, its denominator squared must divide D_0; hence z is integral. Neither V nor y is zero.

Let $F + G\sqrt{D_0}$ be the fundamental positive norm-one unit with integer coordinates. Ordinary Pell classification gives positive indices e_0, m with

$$S = \chi_F(e_0), \quad b = G\psi_F(e_0), \quad x = \chi_F(m), \quad z = G\psi_F(m).$$

Here the sequences are normalized by $(F + \sqrt{F^2 - 1})^n = \chi_F(n) + \psi_F(n)\sqrt{F^2 - 1}$. Because $S \mid x$, we have $\chi_F(e_0) \mid \chi_F(m)$.

The identities

$$\chi_F(j + 2e_0) \equiv -\chi_F(j), \qquad \chi_F(2e_0 - j) \equiv -\chi_F(j) \pmod{\chi_F(e_0)}$$

reduce m to an index $0 \leq j \leq e_0$. Strict growth excludes every $j < e_0$, since $1 \leq \chi_F(j) < \chi_F(e_0)$. Therefore m is an odd multiple of e_0. The multiplication identity implies $\psi_F(e_0) \mid \psi_F(m)$, and $|y| = \psi_F(m)/\psi_F(e_0)$ is integral. □

This is stronger than the generally false inference that a rational number whose square is an arbitrary rational number must be integral. The special square coefficient S^2 is what makes the lemma work.

4.3 Restoration of the other three roots

For a rational t, define

$$C_t = dt^2 + N - e, \qquad U_t = C_t^2 + 4(a_0 - dt^2 N), \qquad \mathcal{E}(t) = U_t^2 - 16C_t^2 a_0.$$

Direct expansion gives the polynomial identity

$$\mathcal{H}_8(dt) = d^4 \mathcal{E}(t). \tag{18}$$

We first justify the denominator bounds without assuming any deleted root. When N_k, N_t are equal units and the twelve supplied variables are positive, weak transport and (11) hold. Since $h \geq 1$, we have $k > E$. Put $A_0 = a + 2$. Then $A_0 < 2E$, $c < 2kY$, and

$$L_0 > A_0 c/4, \qquad L_0 > 4A_0^2, \qquad D < A_0^2 I^2, \qquad b_0 < 2kA_0^2 I^2.$$

For $M_0 = kI^2 + T^2$,

$$M_0^2 \geq 4kI^2T^2, \qquad M_0^2 \geq k^2I^4.$$

Together with $A > L_0^2$, these bounds give $AM_0^2 > 4Dv_0^2$ and $AM_0^2 > 4b_0^2$. For the first inequality it is enough that $k > 16$, which follows already from $k > E$. Splitting into two halves gives

$$AM_0^2 > 2(Dv_0^2 + b_0^2).$$

Since $d = Q - 1 \geq Q/2$ and $M = 2M_0$, we obtain the strict bound

$$N = dM^2A > 4(e + 1) > 0. \tag{19}$$

Suppose now $\mathcal{E}(t) = 0$ for rational t. We have $C_t > 0$. Also $t \neq 0$, because

$$\mathcal{E}(0) = ((N - e)^2 - 4a_0)^2 > 0, \qquad 2\sqrt{a_0} \leq P + Z = e + 1 < N - e.$$

Put $B_0 = 2Qv_0b_0 > 0$. Then $4a_0 = DB_0^2$, and

$$f = -\frac{U_t}{2C_tB_0}$$

is rational with $f^2 = D$, hence an integer. The definition of U_t now yields

$$(C_t + B_0f)^2 = 4d^2t^2M^2A.$$

Thus

$$\tau = \frac{C_t + B_0f}{2dtM}, \qquad y = t - M\tau$$

give an integer τ with $\tau^2 = A$ and initially rational y. Expanding the defining relations gives

$$Q(v_0f - b_0)^2 - (Q - 1)y^2 = 1.$$

Here $v_0f - b_0 = c(Tf - 1) - Rf^2$ is integral, and $Q = (\Delta ic^2)^2$. Lemma 4.2 therefore gives $y \in \mathbb{Z}$. It also follows that $t = M\tau + y$ is integral. The strong equation follows from $f^2 = D$. The restored first and auxiliary roots are nonzero.

4.4 System and single-polynomial soundness

At a three-equation solution, $p > 0$ and $\mathcal{H}_8$ is monic, so the vanishing resultant has a common root. Lemma 4.1 and (17) restore the main norm. Applying the previous argument to $t = v/d$ restores f, τ, y. The signed ordinary interface, using (10), now gives a blank halting computation. One may change both signs (f, T) together and replace τ, y by their absolute values; no sign change is assumed to preserve an expression that depends on an unsquared root.

For the single equation, $N_k, N_t, 1 + \mathcal{R}_{12}$ are integer units. The formula (16) has even h_I, so $\mathcal{R}_{12} \equiv h_R^2 \pmod 4$. Thus $1 + \mathcal{R}_{12} \neq -1$, and $\mathcal{R}_{12} = 0$. The two outer factors have the same sign. The preceding bounds explicitly allowed both signs. After restoration, the inherited signed Pell step-down forces $N_k = N_t = 1$ and $C > 0$. In particular, the exceptional boundary $C = 0$ has not been excluded prematurely.

4.5 Completeness

Start with any canonical positive normalized compiler witness for a finite blank halting run. Set

$$t = M\tau + y, \qquad v = dt, \qquad u = \gamma + cv.$$

All twelve supplied coordinates are positive. The original norm identities give (18) with $\mathcal{H}_8(v) = 0$, and the main norm gives $(W - pv)^2 = B_*$. Hence $\mathcal{R}_{12} = 0$, and all three product factors equal 1. This proves the converse implication.

5 Exact size and the thirteen-variable alternative

5.1 The degree is 1529, not a gate-propagation guess

Give every supplied coordinate degree one and the four coefficients degree zero. The relevant exact degrees are

Expression	c	a	Δ	R	A	I	D	Q, d	P
Degree	5	6	12	4	22	11	34	46	92
Expression	Z, e	a_0	M	N	B_*	W, p	v_R	h_R	h_I
Degree	122	214	23	114	22	11	380	760	581

The unique highest term of v_R is $d^2p^4e^2$; the highest term of h_R is $d^4p^8e^4$. Therefore $\mathcal{R}_{12}$ has the nonzero highest form $d_{\mathrm{top}}^8 p_{\mathrm{top}}^{16} e_{\mathrm{top}}^8$, of degree 1520; the subtracted norm term has degree at most 1184. If $q_0 = \beta J$ and $k_0 = \eta + \zeta$, this highest form is explicitly

$$4^{16} i^{64} k_0^{144} w^{112} s^{256} q_0^{928} (q_0 - F)^{16}.$$

It is nonzero for every authentic $\beta > 0$. The leading forms of N_k and N_t have degrees 7 and 2 and are also nonzero. Their product proves $\deg \mathcal{P}_{12} = 1529$. The emitted circuit costs 73 multiplications and 44 additions/subtractions, totaling 117. Deleting the product finalizer leaves 112 operations for the three equations when the final norm terms are compared directly.

5.2 Retaining the strong root lowers degree substantially

The twelve variables in (2), together with f, give thirteen variables. Retain the common outer definitions, and now put

$$I = ic^2, \qquad K_s = \Delta(f^2 - 1), \qquad d_s = K_s - 1,$$
$$V_f = c(Tf - 1) - Rf^2, \qquad e_s = K_s V_f^2 - 1.$$

The monic quartic

$$\mathcal{H}_4(v) = (v^2 + d_s^2 A - d_s e_s)^2 - 4 d_s^2 A v^2$$

replaces (13). Define

$$\mathcal{R}_{13} = \mathrm{Res}_v(\mathcal{H}_4(v), (W - pv)^2 - B_*).$$

The system is $N_k = N_t = 1$, $I^2 = K_s$, $\mathcal{R}_{13} = 0$, and the single polynomial is

$$\boxed{\mathcal{P}_{13} = N_k N_t (1 + I^2 - K_s)(1 + \mathcal{R}_{13}) - 1.} \tag{20}$$

For an explicit norm evaluator, use $r_0 = W^2 + B_*$ and $r_1 = 2W$, and put

$$s_R = r_0 + p^2 d_s (d_s A - e_s),$$
$$h_R = s_R^2 + B_* r_1^2 - 4 d_s^2 A p^2 r_0,$$
$$h_I = r_1 (2 s_R - 4 d_s^2 A p^2),$$
$$\mathcal{R}_{13} = h_R^2 - B_* h_I^2.$$

The same monic lemma gives integral v and $\gamma = u - cv$. Put $t = v/d_s$. The quartic identity restores $\tau = (d_s t^2 + d_s A - e_s)/(2 d_s t)$ and $y = t - \tau$. Here $t = 0$ is impossible: that branch would give $K_s \mid A - 1$, whereas the weak outer bounds $k > E$, $Y \geq 2$ imply

$$0 < A - 1 = k^2 EY(EY + 1) < k^4 Y^4 < c^4 \leq I^2 = K_s.$$

The rational-square argument restores τ, and Lemma 4.2 restores y, since $K_s = I^2$. The signed ordinary interface proves soundness. Completeness uses $u = \gamma + cd_s(\tau + y)$.

The strong guard cannot be -1: K_s is 0 or 1 modulo 4, as is I^2, so $K_s - I^2$ cannot be 2 modulo 4. The resultant guard has the same even-imaginary-part argument as before. Thus the product has exactly the required existence condition.

The highest form of h_R is $d_s^4 A^2 p^4$, of degree 144. Hence $\deg \mathcal{R}_{13} = 288$, and

$$\deg \mathcal{P}_{13} = 288 + 22 + 7 + 2 = 319.$$

The system costs 83 operations and the product 91. This is a useful intermediate choice when degree matters more than eliminating the last variable.

6 Twenty-six unrestricted integer unknowns

The twelve-variable formula has a larger sound signed domain. Keep exactly

$$J, F, \alpha, z, s, \eta, \zeta$$

positive. Let h, T, u, r, v_w be unrestricted integers and substitute

$$i = 2r + 1, \qquad w = v_w^2. \tag{21}$$

Thus the underlying formula has seven positive and five signed coordinates. Replace each positive coordinate x by its own disjoint copy of

$$\mathrm{Pos}(a_x, b_x, c_x) = 1 + a_x^2 + b_x^2 + c_x(c_x + 1). \tag{22}$$

This is always a positive integer and represents every positive integer. Indeed, the three-square theorem applied to $4x - 3$ gives one odd square and two even squares; dividing the corresponding identity by four gives (22). The resulting polynomial $\mathcal{P}_{26,\mathbb{Z}}$ has

$$\boxed{7 \cdot 3 + 5 = 26}$$

unrestricted integer unknowns. No positivity condition remains outside the equation.

6.1 Why the enlarged domain is sound

The product unit argument still forces $\mathcal{R}_{12} = 0$. Even if $w = 0$, we have $a = Y > 0$, $c > 0$, and $i \neq 0$, so the monic lemma applies first and restores an integral $\gamma = u - cv$ and the main norm. If $w = 0$, then $X = 0$ and reduction of this norm modulo H would give $0 \equiv 1 \pmod{H}$. Hence $w > 0$ and the usual outer bounds follow.

The index equation with a possible unit sign forces $h \geq 0$, since $h \leq -1$ would imply $k \leq R + 1 - E < 0$. Thus $k \geq R - 1$ and $R < 2k$. The old estimate $k > E$, valid in the positive proof, may not be reused here. The following replacement supplies the necessary domination.

The restored main norm classifies $c = \psi_{A_0}(j)$, where $A_0 = a + 2$. Since $c > 1$, $j \geq 2$. The case $j = 2$ gives $c = 2a + 4$ and main root $\pm(2a^2 + 8a + 7)$. Its congruence modulo H forces respectively $X \equiv 4$ or $X \equiv 3a + 2 \pmod{H}$, impossible because $16 \leq X < a < H$. Therefore $j \geq 3$, so

$$c > 4a^2, \qquad k > 2X^2Y, \qquad L_0 > 12A_0^2, \qquad L_0 > A_0 c/4.$$

These imply $A > 144A_0^4$ and $A > A_0^2 c^2/16$. With $M_0 = kI^2 + T^2$, $D < A_0^2 I^2$, and $b_0 < 3kA_0^2 I^2$, one obtains

$$AM_0^2 > 12Dv_0^2, \qquad AM_0^2 > 16b_0^2, \qquad AM_0^2 > 4(Dv_0^2 + b_0^2).$$

This is more than enough for the doubled weight in the actual formula. In particular $C_t > 0$ and $t = 0$ is impossible even when t, T, u are signed.

There is one additional denominator case: $T = 0$. Then $v_0 = 0$ and the eliminant reconstructs $\tau = C_t/(2dtM)$ with $\tau^2 = A$, hence integer τ, and a rational y with

$$Qb_0^2 - (Q-1)y^2 = 1.$$

Lemma 4.2 makes y integral. But $1 < b_0 < Q$: indeed

$$Q - b_0 = (\Delta - R)D - \Delta - c \geq 1 + \Delta(i^2c^4 - 1) - c > 0.$$

There is no integer solution to this norm in that interval. Its rearrangement $(Q-1)(y^2-b_0^2) = b_0^2-1 > 0$ implies $|y| \geq b_0 + 1$ and therefore

$$b_0^2 - 1 \geq (Q-1)(2b_0+1) \geq b_0(2b_0+1) > b_0^2 - 1,$$

a contradiction. Thus $T \neq 0$. All denominators in root restoration are now nonzero, and the same algebra gives integral f, τ, y, γ. Package 14's signed normalized interface applies with precisely its seven positive coordinates, odd i, and square w.

6.2 Completeness and size

Canonical compiler tableaux can have width and height powers of five. Since $d_{\text{base}} = 5^{16}$ and their cyclic length are odd, $q = 2^{d_{\text{base}}N}$ has odd exponent. The recovered rank R is odd, so $w = 2^{R-d_{\text{base}}N}$ is a positive square. Choosing the strong index $m = cR$ yields an odd i, as in the inherited signed-domain proof. This preserves existence; it does not claim every old witness already has these properties. The positive merger witness $u = \gamma + cdt$ is allowed in the larger signed domain. Finally use the surjections (22).

The seven positivity replacements cost seven operations each; odd i costs two and square w one. Thus the product costs $117 + 49 + 3 = 169$ operations, and the system costs 164. The exact degree of the product is 2993; the system's resultant has degree 2976. To see this, assign weight two to the seven positive ports and w, weight one to h, i, T, u, and weight zero to the fixed coefficients. The top degrees are

$$c = 10,\ a = 12,\ \Delta = 24,\ d = 90,\ e = 238,\ p = W = 22.$$

The norm's unique leading form has degree $8(90) + 16(22) + 8(238) = 2976$. Adding the outer degrees 13 and 4 gives 2993. Substitution by the disjoint positive quadratic forms does not cancel this leading form: setting two coordinates in each triple to zero reduces it to an injective squaring substitution.

7 A 59-operation system

There is a small independent improvement for the arithmetic-count objective. Retain the sixteen positive variables

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma.$$

Use the common definitions and put

$$I = ic^2, \qquad K_s = \Delta(f^2 - 1), \qquad V_f = of - c, \qquad U = jc - R.$$

The seven polynomial equations are

$$\begin{aligned} \tau^2 - L_0(L_0 + k) &= 1, \\ (X + ac + H\gamma)^2 - \Delta c^2 &= 1, \\ K_s(V_f^2 - y^2) + y^2 &= 1, \\ k - hE - R &= 1, \\ (K_0 + w)C &= F + z(q-1), \\ I^2 &= K_s, \\ U &= V_f. \end{aligned} \tag{23}$$

The coordinates o and j retain the two auxiliary divisibility conditions; there is no supplied T coordinate in this system. This is the retained ordinary-strong interface proved in package 09. Its circuit is supplied in `data/retained16_system59.json`.

The actual improvement is the coordinate change $z_{\text{old}} = z + 1$. In the old transport equation

$$(K_0 + w)C + (q - F) - z_{\text{old}}(q-1) = 1,$$

this gives $(K_0 + w)C = F + z(q-1)$ and saves one binary operation. At any old positive zero, the compiler gives $w > q > F$ and $C > 0$, so $z_{\text{old}} = 1$ would force $(K_0 + w)C = F$, a contradiction. Hence $z = z_{\text{old}} - 1 > 0$. Conversely every new positive solution gives an old one by adding one. The map is a bijection of positive zero sets.

The emitted system has 33 multiplications and 26 additions/subtractions, for 59 operations. Its seven residual degrees are $22, 18, 24, 7, 2, 22, 6$. A sum of their squares gives a single degree-48 equation using 79 operations. This system is preferable when evaluation cost matters more than the number of supplied integers. It does not improve the 67-operation single polynomial.

8 A smaller fixed coefficient recipe

8.1 Literal machine and cell alphabet

The machine is the accepted dense K215 table, with working states 0 through 214, blank symbol 0, start state 0, and halt state 215. The archive contains its literal table, with SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The actual first action is $(0, 0) \mapsto (1, +1, 93)$. All original actions, including the movement into halt, are retained. Only the padding transitions $(215, b) \mapsto (b, 0, 215)$ are added after halting.

There are 439 cell symbols. Headless bits have codes $0, 1$; a bit b carrying state q has code $2(q + 1) + b$, for $0 \le q \le 215$. Initial phases have codes $L = 434, Q_0 = 435, R_0 = 436$, carrying blank payloads and a head only at Q_0. A vertical wall is $V = 437$ and a horizontal time boundary is $H_0 = 438$. The notation Q_0, R_0, H_0 in this section is unrelated to the arithmetic abbreviations Q, R, H above.

A tile is a four-cell tuple (l, c, r, n), with n the next-time center. Begin with the exact T4 predicate of package 11: at most one head in (l, c, r), propagation and nonadjacency of walls, horizontal propagation of H_0 inside a strip, initialization after H_0, phase edges LL, LQ_0, Q_0R_0, R_0R_0, no head escape, exact deterministic update, and halting before the next H_0.

Strengthen that predicate by the following restrictions:

1. Both neighboring pairs (l, c) and (c, r) obey the horizontal wall/phase restrictions. For a pair not touching V, both entries have the same H_0 status and the same phase status; phase pairs are in the four allowed edges. Across a wall, the adjoining phase must be L on its right and R_0 on its left; two neighboring walls are forbidden.
2. If a horizontal triple contains V, all its symbols belong to $\{0, L, R_0, V, H_0\}$. Thus nearby ordinary cells are blank and headless.
3. If $n = H_0$, none of l, c, r is an initial phase and every head in the triple is in halt state.
4. If $c = H_0$ and $l = V$, require $n = L$; if $r = V$, require $n = R_0$.

There are exactly 5,251 tuples satisfying the strengthened rule. Add the old locally valid but globally impossible tile $E_0 = (H_0, 0, 0, H_0)$, and order the alphabet as

$$S_0 = (L, Q_0, R_0, 1), \quad E_0, \quad \text{all remaining strengthened tuples in lexicographic order.}$$

The resulting list has 5,252 entries and is supplied in `data/strict_windows.json`. It is a finite list computed from the 430 literal actions; generating it does not search for a K215 halt.

8.2 Why the stricter language preserves the target computation

Every tile in the new list is an old T4 tile. Therefore any periodic new tableau gives an old one, and the established marked-strip argument proves soundness. The forbidden tile E_0 cannot occur globally: its left neighbor is H_0 with a nonwall right neighbor 0, contradicting the horizontal boundary rule at that left cell. It is retained only as the selector that the arithmetic mask excludes.

For completeness, pad a finite genuine halting run with at least two unvisited blank cells next to each wall. Such cells remain blank throughout. Every ordinary triple touching a wall is then blank and headless; initial rows use L and R_0 at the corresponding margins, and boundary rows use H_0. All other new restrictions hold on genuine rows. In the helical periodic presentation, a wall separates neighboring strips whose time origins differ by one. The allowed wall alphabet includes every combination of their possible blank, phase, and boundary edge symbols, so these seams also satisfy the restrictions. Independent width and height padding remain available, including arbitrarily large powers of five.

The original three overlaps are unchanged:

$$w_i(r) = w_{i-1}(c), \qquad w_i(c) = w_{i-1}(l), \qquad w_i(n) = w_{i-h}(c).$$

They reconstruct the plane from the center word even when residues collide. No condition $h \mid N$ or $\gcd(h, N) = 1$ has been added.

8.3 Exact four integers

Set

$$a_c = 439,\ k_c = 5252,\ \ell = 13,\ A_c = 2^{13},\ \rho = 2^{78125}, \quad B_{\text{base}} = \rho^{1953125} = 2^{5^{16}}.$$

For $0 \le j \le 438$, the four copy positions are

$$e_l(j) = 8764 + j, \quad e_c(j) = 8325 + j, \quad e_r(j) = 7886 + j, \quad e_n(j) = 7008 + j.$$

Let the native support $\mathcal{S}$ consist of selectors $0, \ldots, 5251$, these four copy intervals, dummy positions $10520, 10521$, and anchors

$$M_0 = 11839, \quad 3M_0 = 35517, \quad 5M_0 = 59195, \quad 11M_0 = 130229.$$

Thus $|\mathcal{S}| = 7014$. Set

$$\mu = A_c - 2 + \sum_{j=1}^{2835} A_c^j.$$

For tile $w_i = (l_i, c_i, r_i, n_i)$ define

$$v_i = 1 + A_c^{1+l_i} + A_c^{440+c_i} + A_c^{879+r_i} + A_c^{1318+n_i} + \sum_{t=1}^{4} A_c^{1756+t},$$

$$v_{e_l(j)} = A_c^{1+j}, \qquad v_{e_c(j)} = A_c^{440+j},$$

$$v_{e_r(j)} = A_c^{879+j}, \qquad v_{e_n(j)} = A_c^{1318+j}.$$

At anchors $M_0, 3M_0, 5M_0, 11M_0$, respectively, use $A_c^{1757}, A_c^{1758}, A_c^{1759}, A_c^{1760}$. Both dummy coefficients are zero. Put

$$\begin{aligned} D_C &= \rho^{1317} + \rho^{226698} + \rho^{47356} + \rho^{94712} + \sum_{e\in\mathcal{S}} v_e(\rho^{487157-e} + \rho^{747619-e}), \\ D_R &= \rho^{226259}, \\ M_F^0 &= \mu(\rho^{487157} + \rho^{747619}) + \sum_{j=0}^{438} \rho^{8325+j} \\ &\quad + \sum_{j=0}^{438} (\rho^{226259+8764+j} + \rho^{226259+8325+j}) + \rho^{59195} + \rho^{130229}, \\ M_C &= B_{\text{base}} - 1 - \sum_{\substack{e\in\mathcal{S}\\ e\neq 1}} \rho^e - 2\rho^{10520}, \qquad M_F = M_F^0 + 4. \end{aligned} \tag{24}$$

The four coefficients are now exactly

$$\boxed{\beta = B_{\text{base}} - 1, \quad K_0 = D_C + B_{\text{base}} D_R, \quad M_C, \quad M_F^* = M_F + B_{\text{base}} - 1.}$$

All exponentiations in this recipe have fixed integer exponents. These constants are independent of the unknowns and of whether K215 halts.

8.4 Compiler geometry and population proof

The new anchor set is $\{M_0, 3M_0, 5M_0, 11M_0\}$. The tested temporal anchor targets are $5M_0, 11M_0$, reached by bare shifts $4M_0, 8M_0$ from the first two anchors. The only support pair with difference $6M_0$ is $(5M_0, 11M_0)$: nonanchor support is below M_0, and the other anchor differences are different. The circumference is larger than twice the support diameter, so there is no wrapped alternative. This replaces the older anchor set $\{M_0, 3M_0, 9M_0, 27M_0\}$.

The remaining scalar layout is

$$E_{\max} = 130229, \quad H_c = 226259, \quad T_1 = 487157, \quad T_2 = 747619, \quad L_c = 1953125.$$

All copy/anchor targets lie below $T_1 - E_{\max}$. At each target, the four bare shifts in D_C hit exactly the intended source positions, and $D_R = \rho^{H_c}$ tests exactly the intended spatial copies. There are 1,319 such targets. The supplied finite set checks exhaust all of them. The clause bands have separation $T_2 - T_1 = 260462 > 2E_{\max}$, while the opposite circular separation is also larger than $E_{\max}$. Thus every rotation leaves at least one clean clause band.

All bits cleared from M_C are distinct and some highest bits remain; all fields of M_F lie strictly below bit 5^{16} and leave unused high bits. Thus $0 < M_C, M_F < B_{\text{base}} - 1$. The selector at position zero clears bit zero of M_C, while bit one remains. The only contribution to M_F below the first radix position is the marker 4. Therefore $M_C \equiv 2 \pmod 4$ and $M_F \equiv 4 \pmod 8$.

The mask population is exact:

$$\text{pc}(\mu) = 12 + 2835 = 2847, \quad \text{pc}(M_F) = 2(2847) + 3(439) + 2 + 1 = 7014, \quad \text{pc}(M_C) = 5^{16} - 7014.$$

The last $+1$ in $\text{pc}(M_F)$ comes from the native low-bit marker 4. There are two low dummy positions, both with zero clause coefficient. Only the first receives the extra allowed upper bit through $-2\rho^{10520}$ in M_C. Hence there is still only one exceptional upper spill digit in the all-rotation proof. The two anchor tests exclude every nonzero binary rotation within a radix digit; the unique anchor difference then excludes every nonzero radix rotation within a cell.

The carry bounds are also checked with manageable integers: μ has bit length 36,856, the sum of clause coefficients has bit length 22,893, and the required mass threshold has bit length 22,909. The chosen radix exponent 78,125 is larger than all of these bounds, and the native compiler's carry-free estimates apply.

Finally, $b = 78125$ is odd, so $\rho \equiv 2 \pmod 5$. The band separation is 2 modulo 4, so paired clause contributions in D_C cancel modulo 5. The two new bare anchor shifts are multiples of 4, and $439 \equiv 3 \pmod 4$. Therefore

$$2D_C - D_R \equiv 3 \pmod 5.$$

This is the unit required to tune the actual packed rank using the first dummy bit. Width, height, and d_{base} remain powers of five. The original dummy-control completeness and canonical positivity margin therefore apply unchanged. Together with the local-language proof, this establishes the complete four-port interface for the new tuple.

9 Metatheory, verification, and integration

9.1 The logical endpoint

The existing K215 endpoint is

$$\text{Halt}(K215, \text{blank}) \longleftrightarrow \text{there is a finite ZFC derivation of falsity.}$$

Negating this and composing with the proved Diophantine equivalence gives Theorem 1.1. No consistency premise is used to prove the biconditional. If ZFC is inconsistent, both a proof-search halt and a finite integer witness exist. If it is consistent, the corresponding equation has no solutions in the stated domain.

The distinction between this statement and a proof of consistency is substantive. Under the usual second-incompleteness hypotheses, consistency prevents ZFC from proving the no-solution statement. Mere consistency alone does not justify a claim of two-sided unprovability; an appropriate soundness assumption is needed for the stronger claim. No two-sided independence claim is required here.

For ordinary nonnegative natural-number unknowns, replace each positive coordinate of $\mathcal{P}_{12}$ by $x + 1$. This preserves the number of unknowns and the degree. It adds twelve operations to the literal evaluator. Do not interpret the original positive polynomial directly over all of $\mathbb{Z}$; Section 6 supplies the separately proved unrestricted form.

9.2 What the checks establish

The archive contains division-free JSON circuits, standard-library Python emitters, and recorded checks. The checks establish the live operation counts, algebraic identities, degree bounds and leading-form witnesses, agreement with an independent quadratic-pair evaluation, the literal machine/table binding, exact finite tile enumeration, field positions, carry bounds, anchor synchronization, and padded control-tableau seams.

The checks do not establish Con(ZFC), do not produce an accepting K215 run, and do not replace the unbounded proofs in this report. The enormous coefficients are never expanded. No fresh end-to-end Lean or Rocq build was performed. Independent reviews of the new monic elimination and strict compiler are included in `proofs/`.

9.3 Reproduction and file layout

Run the included reproduction driver with Python 3.10 or later. It uses only the standard library for the required arithmetic and finite-data checks. Compile this report with pdfLaTeX twice, or with the provided Makefile. The top-level README gives the exact commands and integration destination. The archive is self-contained for its new checks and includes the inherited mathematical interfaces used in the proofs.

The recommended integration is a new numbered synthesis directory under the existing `con-zfc-diophantine-forms` report collection. Keep the earlier packages as comparison points for objectives they still optimize. In particular, retaining the 67-operation polynomial is appropriate even though its coefficient tuple can be rebound to the new strict compiler.

9.4 Remaining work

A useful next formalization target is Lemma 4.2, followed by Lemma 4.1 and the monic identities. These are relatively small arithmetic results and isolate the genuinely new proof obligations. The larger task remains formalization of the native-window/Pell bridge. Further variable elimination would need a new integrality argument for another quotient; simply clearing its denominator or taking another resultant would not establish the required equivalence.

References

[1] V. Reshetnikov et al., *ProveIt*, snapshot `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`, 22 October 2026. Repository paths are given in the package's provenance manifest and in `reference/comparison_at_pin.md`.
https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt, *Con(ZFC) in Diophantine form*, wave-2 packages 09–15, at the pinned snapshot. Principal dependencies: 09 weighted roots; 11 T4 compiler; 12 ordinary signed interface; 14 signed-domain reduction.

[3] ProveIt, *Formal correctness of the exact 215-state machine*, and *candidate-k215/proof.md*. The retained endpoint names are `KEndpoint215.blank_halts_iff_zfc_refutation` and `tm215_halts_iff_ZFC_inconsistent`.

[4] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571. https://doi.org/10.2307/2273588.

[5] Yu. Matiyasevich and J. Robinson, *Reduction of an arbitrary Diophantine equation to one in 13 unknowns*, Acta Arithmetica **27** (1975), 521–553. https://doi.org/10.4064/aa-27-1-521-553.

[6] J. Bayer, M. David, M. Hassler, Yu. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909v2 (2025). https://arxiv.org/abs/2506.20909.